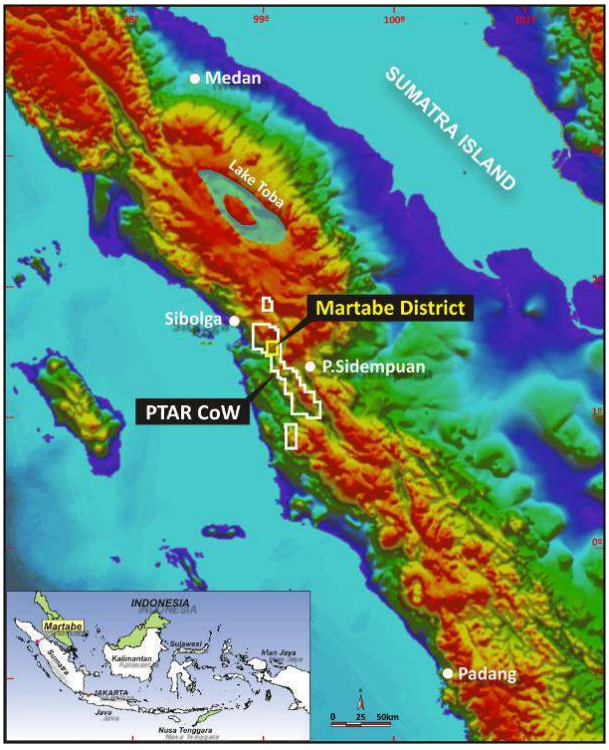
Geological Resume of the Martabe (Especially Ramba Joring) Deposit
Martabe Gold Mine - PTAR
North Sumatra, Indonesia

The Martabe gold district is located 30 km to the east of Sibolga on the West coast of Sumatra - Indonesia. Exploration activities have been conducted in this area by several Companies since 1997. PT Agincourt Resources is now the operator of the Contract of Work (CoW) at Martabe, Kecamatan Batangtoru, Kabupaten of Tapanuli Selatan. All deposits are located outside of "Protected Forest" (Hutan Lindung). The Contract of Work (CoW) area lies between 1° and 2° north of the equator and experiences a typical wet tropical climate at elevations up to 600m, with a monsoonal rainfall between October and April and a rainfall averaging 4.5 m/year. The temperature ranges from 15 to 32°C. It is a 6th generation of CoW that was issued by the Ministry of Mine and Energy of Indonesia on March 1997.

Figure 1: Sibolga Cow Location



Sumatra is located in front of the subduction zone between the the Indian-Australian plate and the Sunda craton. Indian-Australian plate is being subducted along the Sunda Trench at rate of 80 mm/year. As result the Sunda volcanic arc developed parallel to the trench and forms the Barisan mountain ranges.

Wrench fault system has been developed along the arc, which is being active and is characterised by volcanism associated with high levels hydrothermal fluid flow, which occurs with both extensional and compressive regimes, consistent with a wrench system.

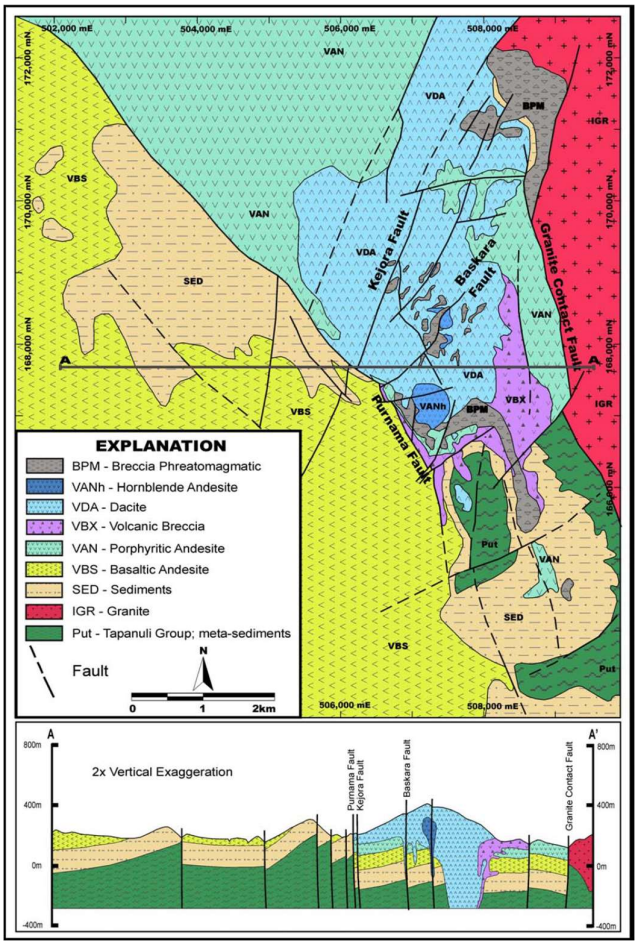
The Martabe district lies on the western flank of the Barisan Mountains. This area is located west of the northwest-striking Axial Barisan Zone, which is the geomorphic expression of a number of fault-bound grabens within the Sumatran Fault Zone (SFS) where the most recent fault activity is focused.

I. Stratigraphy of Martabe

The central part of the Martabe district is characterized by steep fault-bounded volcanic hills and sedimentary escarpments. Sedimentary rocks occur in a number of locations in the district, and their stratigraphic position is problematic.

Mapped lithologies are provisionally divided into five major units (from oldest to youngest):

Figure 2: Geological map and cross-section of Martabe District



- Pre-Tertiary carbonaceous meta-sediments of the Tapanuli Group (Put)
- Pre-Tertiary granite assigned to the Sibolga Granite Complex (IGR)
- Tertiary Barus Formation composed of sandstone and conglomerate (SED)
- Tertiary Angkola Volcanics of basalt-andesite composition (VBS, VAN, VBX)
- Late-Tertiary dacite-andesite dome and diatreme complex (BPM, VDA, VANh)

Tapanuli Group and Sibolga Granite Complex

The oldest rocks in the Martabe district belong to the Permian-age Tapanuli Group (Put) comprising a gray to black meta-graywacke and shale-slate sequence of unknown thickness. Metamorphic effects are generally weak but variable and original lithologies include carbonaceous siltstone, mudstone, fine-grained sandstone and minor grit-conglomerate that locally exhibit graded bedding, small scale ripple laminations and are locally calcareous. Other textures include flaser structures and slaty cleavage. The Tapanuli Group is interpreted to be a submarine turbidite deposit that has been subjected to low-grade metamorphism.

The eastern part of the district is dominated by granite (IGR) that forms both intrusive and fault contacts with the Tapanuli Group. Based on Ar-Ar age dating the Martabe granite is 209Ma (Turner, 2004) or approximately the same age as the Sibolga Granite Complex that forms a batholithic size intrusion complex 20 to 40 kilometers northwest of Martabe.

Tertiary Sedimentary Lithologies

Conglomerate, sandstone (sometimes micaceous), and siltstone with minor limestone and carbonaceous shale are mapped in the central and eastern parts of the district and are tentatively assigned to the Barus Formation (SED) of late Oligocene to middle Miocene age. Fossils from sandy limestone collected below the Angkola Volcanics are interpreted to be 12-25 Ma, early to mid-Miocene (Lunt, pers. com., 2001). Cross-bedded and massive textures are most common. Drill data and regional mapping (Aspden et al., 1982) indicate the Barus Formation underlie most of the Martabe district.

Coarse-grained sandstone and conglomerate are well exposed and forming prominent cliffs that correspond to major air photo features. These rocks are strongly silicified and therefore resistant whereas Barus Formation elsewhere in the district tends to be friable and weakly indurated. Siltstone and sandstone crop out in the western part and core drilling intercepted similar lithologies within the diatreme complex. These rocks are interpreted to be rafted blocks of Barus Formation although it is likely that at least some of these sedimentary beds, particularly material found at higher elevations in the diatreme complex, are lake sediments deposited in the diatreme depression and preserved as subsidence blocks.

Miocene Volcanic Lithologies

Overlying the Barus Formation is a series of basaltic and andesitic lavas of unknown thickness belonging to the middle Miocene Angkola Volcanics. In the Martabe district, the Angkola Formation is divided into three lithologies: fine grained basalt (VBS), fine- to medium-grained porphyritic andesite (VAN) and volcanic breccia of andesite composition (VBX). The fine-grained basalt is the lower most unit of the Angkola in the Martabe area and it is widespread west of the fault; however, little is known about this area due to the lack of significant alteration and exploration potential. Basalt (VBS) has recently been dated using whole rock Ar-Ar at 18.91 Ma (Turner, 2004). Porphyritic andesite (VAN) contains about 20-25 percent 5-12 mm plagioclase phenocrysts and is interpreted to be a lava flow. The volcanic breccia (VBX) is a coarse heterolithic fragmental rock composed primarily of subrounded clasts of VAN but locally contains sandstone and siltstone fragments. Initially the VBX was interpreted to be a volcaniclastic autobreccia, while the current interpretation is that it is a product of early phreatic explosion peripheral to the main diatreme complex. Primary copper mineralisation is dominated by high-sulphidation assemblages of enargite and covellite and in the unoxidised zones is associated with similar acid sulphate alteration to the gold-bearing zones. The distribution of copper is significantly complicated by oxidisation affects, with leaching apparent in strongly oxidised zones and supergene enrichment apparent in transitional zones.

The coincidence of broad northeast strike and sub-vertical dip of breccia zones, advanced argillic alteration zones and massive to vuggy silica alteration is considered strong evidence that a relatively simple structural regime operated at the time of mineralization, creating dilation in these northeast structures allowing fluid assisted brecciation, alteration and mineralization.

The mineralization in Ramba Joring is distinguished into two types – Sulphide and Oxide mineralization.

Late Tertiary Dacite Porphyry and Hornblende Andesite

Emplaced along the fault dissected western margin of the granite complex is a north-trending exogenous dacite dome field. Two closely related lithologies are recognized: dacite porphyry (VDA)

and hornblende andesite (VANh). The Dacite is predominantly porphyritic with 20-30 percent plagioclase crystals, 5-10 percent biotite and hornblende and 3-7 percent quartz phenocrysts. Textures include flow banding, autobreccia and local fine-grained pyroclastic rocks having the same composition (Levet, 2003).

The hornblende andesite (VANh) is a porphyritic rock with 10-15 percent distinct 8mm long hornblende phenocrysts, 15-20 percent plagioclase feldspar grains and rare quartz (<2 percent) phenocrysts. Xenoliths of fine-grained material of the same composition are common and flow banding and grain alignment occurs locally.

Late Tertiary Phreatic and Phreatomagmatic Breccia

Closely associated with dome is a diatreme complex and series of fault-controlled diatreme breccia dikes. The general term Breccia Phreatomagmatic (BPM) is used to describe a large variety of breccia types of both phreatic and phreatomagmatic origin. However, because of their close association with hydrothermal alteration and mineralization a silicified Breccia Phreatomagmatic (sBMP) and clay-rich Breccia Phreatomagmatic (cBPM) variety are distinguished when appropriate.

The dacite porphyry, hornblende andesite, and BPM breccias have not been dated but are assumed to be late-Tertiary and probably Pliocene age based on cross-cutting relations suggesting they are closely linked in time to the epithermal alteration (alunite) dated at 2.1 to 3.3 Ma (Turner, 2004).

Quaternary Toba Tuff

The Quaternary Toba Tuff forms a prominent plateau north of Martabe while in the district it is restricted to small, isolated erosional remnants, in the district geology map. The Toba Tuff consists mostly of unwelded rhyolite tuff with variable pumice and lithic clasts.

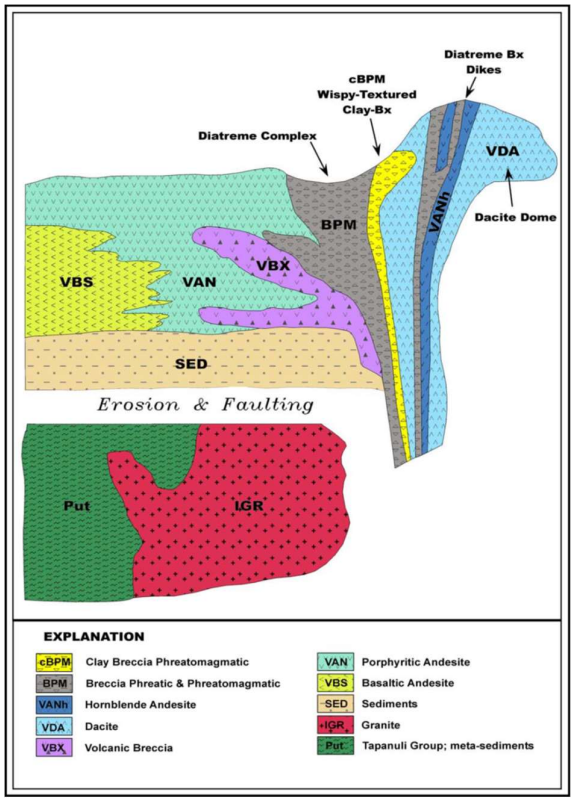


Figure 3: Generalized stratigraphic section of Martabe District

II. Structure of Martabe

The Martabe district flanks an uplifted block of Permian meta-sediments and Permo-Triassic granite to the east. Three principal fault orientations are interpreted from lineaments and recognized from mapping: northwest, north-northwest, to north-northeast and east-northeast trending. Davies (2002) suggests that each of these structural orientations have equivalents at a larger regional scale. Observations and field data provide sufficient constraints to allow these structures to be interpreted in the context of a kinematic model.

Map data suggests northwest-striking faults show right lateral strike-slip displacement and are interpreted to represent strands of the Sumatra Fault System (SFS). A complex strain environment is indicated by a significant component of vertical displacement as well as lateral offset. The downthrown north block of the fault preserves the diatreme complex and high-sulfidation epithermal deposit at a high-level of exposure.

The second fault set consists of a series of north-northwest to north-northeast faults and fracture zones that form a structural graben in the north central part of the Martabe District. The most important of these structures is the Granite Contact fault which marks the contact between granite to the east with volcanic, intrusive and sedimentary rocks of the Martabe volcanic domain to the west. North-striking faults and fractures are interpreted as controls for emplacement of the dacite dome complex within the graben. Mapping and drilling along the Granite Contact fault has encountered fault talus, or fault scarp breccia which indicate normal movement with the downthrown block to the west (Davies, 2002). Hydrothermal alteration centers are also aligned along the north trend, suggesting that structural breaks with that orientation are a major ore controlling feature.

East-northeasterly striking faults are less dominant than the other two structural orientations described above, however they are important elements in understanding the overall tectonic framework and are interpreted to be compressional release mechanisms with a component of reverse fault displacement (Davies, 2002). Uplifted blocks are generally to the southeast with fault planes dipping towards the up thrown blocks (SE). They do not appear to significantly control the distribution of alteration or mineralization and therefore there is little drill data to better document their character.

III. Lithology of RJ Deposit

Coherent Volcanic (Porphyritic Dacite-Andesite)

The dominant volcanic lithofacies at Ramba Joring is coherent feldspar porphyry of dacitic to andesite composition, probably emplaced as a high-level sub volcanic intrusion. Generally, is characterized by light brown to grey colour, porphyritic, white tabular phenocryst of feldspar (~15-20%, 2-6mm), elongate prismatic amphibole (hornblende? 5-10% 1-4 mm), euhedral to rounded commonly partly resorbed quartz phenocryst (~2-10%, 1-4 mm), and locally sparse flakes of biotite (<2%, < 2mm) evenly dispersed, sometimes weakly aligned, in a pale grey originally feldspathic ground mass (Herman,2010).

The similarity of feldspar and quartz phenocryst in both end member, VANh and VDA, suggests a close magmatic affinity, possibly a continuum between them, perhaps indicating some fractionation or primary in homogeneity in the source magma chamber.

No convincing evidence for intrusion of porphyry into breccia, or autoclastic fabrics have been observed that might suggests eruption and emplacement at surface. Therefore, the porphyry is interpreted as having been emplaced as coherent sub volcanic high-level porphyritic intrusion or intrusive complex.

Phreatomagmatic Breccia (BPM)

Phreatomagmatic breccia at Ramba Joring is unstratified breccia and characterized by whitish brown to dark grey color, monomict to polymict, angular to sub angular clasts of mostly porpyritic Dacite-Andesite clasts with minor sandstone-siltstone and quartz clasts, medium to very coarse-grained, matrix supported with sandy-silty carbonaceous matrix. Generally this rock has altered to silica with vuggy texture and envelopad by advance argillic alteration (silica-alunite-dickite-kaolinite) and minor clay (kaolinite/dickite-illite) alteration.

Based on descriptive classification, there are 4 (four) different facies in Ramba Joring; Unstratified breccia sandy matrix monomict-polymict lithic clasts (XUS), Unstratified white/black quartz matrix breccia (XUQ-w/b), unstratified sulphide matrix breccia (XUQ-ms) and unstratified iron oxide matrix breccia (XUIO).

The unstratified sandy matrix monomict-polymict lithic clast breccia (XUS) is interpreted as the phreatomagmatic breccia, whilst the other three (3) – XUQ-w/b, XUQ-ms and XUIO, are classified as hydrothermal breccias.

Hydrothermal breccia (BHX)

The breccia also is typically affected by alteration, they most commonly lie in alunite facies altered zones. That is consistent with the idea that primary permeability of the breccia allowed greater hydrothermal fluid flow than coherent porphyry.

There are three different facies of hydrothermal breccia are distinguished in Ramba Joring prospect, based on its matrix; unstratified white/black-quartz matrix breccia (XUQ-w/b), unstratified massive sulphide breccia (XUQ-ms) and unstratified iron oxide matrix breccia

IV. Alteration of RJ Deposit

The alteration mineral assemblages were determined by ASD Teraspec™ spectral to oversee the detail hydrothermal alteration mineral composition (especially in clay minerals) and visual logging of quartz (silica) alteration that already existed in numerical form in drill hole geological database.

A total of 46.765.1m drill cores and 725 surface samples were analysed with ASD Teraspec™ spectral. Based on the ASD spectral characteristics, at least six (6) mineral assemblage zonation are identified within Ramba Joring deposits that represented a zonation of different temperature and acidity of the hydrothermal fluids.

The alteration mineral assemblages at Ramba Joring describe a classic acid sulfate epithermal hydrothermal system ranging from mildly acidic low temperature illite facies in the outer parts, through an advance argillic series of halloysite, kaolinite, dickite, pyrophyllite and alunite associated with progressive increase in acidity and temperature (Herman, 2010).

Silica (SI)

The silica alteration zone varies from massive quartz, vuggy quartz to chalcedony/opal, but vuggy quartz is a more common silicification at Ramba Joring deposit. The degree of silicification is varying, starting from moderate, strong and very strong. Quartz frequently associated with pyrite, is an indicator of high hydrothermal fluid flow zones at Ramba Joring, in both vuggy texture zones and in subsequent zones of silicification in previously leached vuggy zones. The high gold mineralisation in Ramba Joring is mostly associated with late-stage silicification.

Silica alteration mostly occurs at the western part of Ramba Joring deposit with NE trending and at Southern part with E-W trending. The silica alteration zones at Ramba Joring deposit is influenced by secondary order of Sumatra fault system.

Quartz ± Alunite ± Pyrophyllite (Strong Advance Argillic)

This alteration facies are predominantly characterized by quartz-alunite assemblages with minor pyrophyllite. Quartz ± alunite facies in the Ramba Joring drill core typically have pronounced vuggy texture, suggesting extreme leaching, which is turn implies high hydrothermal fluid-rock ratio. The quartz – alunite facies commonly also associated with gold mineralisation.

Alunite occurs in both the breccias and porphyritic volcanic andesite dacite, which is replacement masses of phenocryst in porphyritic volcanic andesite dacite and in clast and matrix of breccias.

Dickite ± Kaolinite (Weak Advance Argillic)

Dickite ± kaolinite alteration facies occur peripheral to or on the outer margins of quartz – alunite alteration where gold grade mineralisation is mostly low. This alteration facies is predominantly characterized by dickite – kaolinite assemblages with minor illite – alunite - quartz

Kaolinite ± Illite (Argillic)
This alteration facies are predominantly characterized by kaolinite – illite - smectite assemblages with minor illite-quartz. Kaolinite - illite alteration facies occur in peripheral to both the quartz – alunite and dickite – kaolinite alteration.

Chlorite ± Calcite (Propylitic)
Propylitic alteration constitutes the outer parts of the Ramba Joring deposit and characterized by chlorite – calcite and minor epidote. The alteration zone is evident in the eastern part of Ramba Joring deposit.

V. Oxidation of RJ Deposit

Oxidation stage in Ramba Joring has a complex and highly variable pattern of oxidation, with the base of complete oxidation surface plunging down and situated adjacent to un-oxide material due to fractured and faulted. Hematite, goethite, kaolinite and supergene alunite after sulphate and sulfosalt are the most common product of oxidation.

The oxidation in Ramba Joring is variable, ranging from 40m to 100m and deeper within faults or fractures as mentioned above. The distribution of gold in Ramba Joring is mostly set in oxide zone but still present in transitional and un-oxide (sulphide) zone.

VI. Mineralization of RJ Deposit

Gold mineralisation at Ramba Joring is interpreted to occur in a series of northeast striking, sub-vertical zones that are principally defined by the combination of advanced argillic alteration (defined as silica-alunite or massive to vuggy silica) and gold grade. The zones are partially coincident with breccia zones, however the breccia zones are not interpreted to be a primary control on mineralisation as strong gold grades and alteration do occur in coherent rock (ie outside breccia zones).

Primary copper mineralisation is dominated by high-sulphidation assemblages of enargite and covellite and in the unoxidized zones is associated with similar acid sulphate alteration to the gold-bearing zones. The distribution of copper is significantly complicated by oxidisation affects, with leaching apparent in strongly oxidised zones and supergene enrichment apparent in transitional zones.

The coincidence of broad northeast strike and sub-vertical dip of breccia zones, advanced argillic alteration zones and massive to vuggy silica alteration is considered strong evidence that a relatively simple structural regime operated at the time of mineralization, creating dilation in these northeast structures allowing fluid assisted brecciation, alteration and mineralization.

The mineralization in Ramba Joring is distinguished into two types – Sulphide and Oxide mineralization.